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Natural Language Processing

Assignment 1

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1 Dataset & Setup

This project focuses on the AGNews Dataset, created by Zhang et al. [1]. The dataset consists of 120k training datapoints and 7.6k testing points. Each datapoint consists of a title and a description of a news article. These are around 45 and 200 characters long on average. Each article is categorised based on topic. The set of labels consists of World, Sports, Business and Sci/Tech ($C = 4$, $y \in \{1, 2, 3, 4\}$). This project aims to investigate the performance of baseline models on this task. For hyperparameter tuning and testing, a random development split of 10% was taken from the training set.

To train models, first, the data needs to be preprocessed. The first step was to concatenate the title and the description into a single string of text, which makes it easier to use both features.

Next, an embedding needs to be made, his was done using a Term Frequency Inverse Document Frequency (TF-IDF) Vectorizer as implemented in Scikit-Learn [2]. The term frequency of a word is the number of times it occurs in a given document (sample). The inverse document frequency is a measure of how much information a term provides; this is based on the number of documents in which the term appears. The idea is that terms that appear in fewer documents provide more information than those that appear in many documents. Using these two steps, a co-occurrence matrix is computed for each document, and the different entries are weighted based on the TF-IDF values. For this project, it was chosen to work with 5000 features. Thus for each sample i we get a sparse matrix $x_i \in \boldsymbol{M}_{5000\times5000}$. Additionally, we chose 33 as our fixed random seed.

2 Models

2.1 Logistic Regression

Logistic Regression consists of a linear classification model, as specified in Equation (1)

$$\boldsymbol{a} = \boldsymbol{W}\boldsymbol{x} \tag{1}$$

where $\boldsymbol{a}$ is a vector of logits, $\boldsymbol{x}$ is the feature vector and $\boldsymbol{W}$ is a matrix of weights [3]. To convert these logits to a probability, the softmax activation function is used, specified in Equation (2) [3].

$$p(y = c|\boldsymbol{x}, \boldsymbol{W}) = \frac{e^{a_c}}{\sum_{c'=1}^{C} e^{a_{c'}}} \tag{2}$$

The final prediction of the class is the value for which this probability is maximal. The model is trained through Maximum Likelihood Estimation (MLE), in which it finds the parameter values that make the observed data most probable. To solve the optimisation problem, the Limited Memory BFGS algorithm is used, in which we approximate the curvature from the second derivative to calculate the minimum error. By default, the model from Scikit-Learn applies L2 Regularisation [2]. This modifies the objective function by adding a penalty term proportional to the square of the weight magnitudes controlled by the inverse regularisation strength hyperparameter C.

2.2 Support Vector Machine

The goal of a Support Vector Classifier (SVC) is to find a maximum separation hyperplane between data points of two sets [4]. To generalise this to more than two classes, a one-vs-rest classification strategy was used. A hyperplane in an inner-product space is specified using the orthogonal vector $\boldsymbol{w}$ and an offset b. To measure this, we require all data points to be members of an inner-product space, which is $\boldsymbol{M}_{5000\times5000}$ in the present case. We calculate the error metric with Hinge Loss, seen in Equation (3) [4].

$$L = \max(0, 1 - y_i(\boldsymbol{w} \cdot \boldsymbol{x}_i + b)) \tag{3}$$

Since there is not always a hard margin between two sets, to find the optimal separating hyperplane, it is necessary to use a slack on the loss. The ratio between minimising the training error and maximising the margin is parameterised by C [4]. This is achieved by minimising the objective function, specified in Equation (4)

$$\min_{w,b,\zeta} \frac{1}{2}||w||^2 + C\sum_{i=1}^{n}\zeta_i \tag{4}$$

where ζ_i reflects the the Hinge loss for each datapoint and C is the (tuned) hyperparameter [4]. The penalty ensures that points falling inside the margin or on the incorrect side of the hyperplane are penalised linearly. Unfortunately, there is not a consistent way to select this hyperparameter a priori; thus, for this project, it was opted to tune this using a 5-fold grid search on $C \in \{10^{-3}, 10^{-2}, 10^{-1}, 10^{0}, 10^{1}, 10^{2}, 10^{3}\}$. Based on these results, $C = 10^{-1}$ was the best option. The full results can be found in Appendix F.

3 Experimental Protocol

The main goal of this report is to compare the baseline models above. For these purposes, we train both models on the same training set. We evaluate the models on the development and testing sets based on multiple metrics. The models are evaluated based on accuracy, weighted F1-score (which accounts for label imbalance) and a confusion matrix. Additionally, an elaborate error analysis is done for the best model, focusing on the different types of errors and their (potential) causes.

4 Results

4.1 Logistic Regression

The Logistic Regression model reported, for the development set, an accuracy of 0.8748 and a weighted F1-score of 0.8742; for the test set, the model reported an accuracy of 0.7470 and a weighted F1-score of 0.7455. The model achieved a higher performance on the development set but saw a noticeable decline when applied to the test set. The confusion matrix comparison is shown in Table 1.

4.2 Support Vector Machine

The SVM model, tuned with $C = 0.1$, reported, for the development set, an accuracy of 0.8776 and a weighted F1-score of 0.8769; for the test set, the model reported an accuracy of 0.7661

	Dev Set				Test Set			
Pred \ Act	**World**	**Sports**	**Business**	**Sci/Tech**	**World**	**Sports**	**Business**	**Sci/Tech**
World	2539	189	146	102	1520	152	101	127
Sports	52	2694	23	20	129	1588	67	116
Business	169	22	2584	264	222	133	1235	310
Sci/Tech	119	64	333	2680	196	122	248	1334

Table 1: Logistic Regression Confusion Matrix Comparison

and a weighted F1-score of 0.7645. This model slightly outperforms the Logistic Regression model on both the development and test sets, and demonstrates better robustness. The confusion matrix comparison is shown in Table 2.

	Dev Set				Test Set			
Pred \ Act	**World**	**Sports**	**Business**	**Sci/Tech**	**World**	**Sports**	**Business**	**Sci/Tech**
World	2515	202	166	93	1542	143	88	127
Sports	36	2719	16	18	101	1632	52	115
Business	146	23	2598	272	203	133	1251	313
Sci/Tech	128	59	310	2699	192	125	186	1397

Table 2: SVC Confusion Matrix Comparison

5 Error Analysis

5.1 General

Looking at the confusion matrix of the Logistic Regression shows us that sports is the least often misclassified of all categories. Business, on the other hand, is most often misclassified. The most common misclassifications are Business being classified as Sci/Tech and Sci/Tech being classified as Business. This could possibly be explained by the fact that a lot of Business articles contain words and sentences like: "The launch of ...", "The largest Internet Retail ...", "The release to manufacturing of Microsoft Operations Manager". In short, in the misclassified articles that fall in the Business class, words appear that can signify an article falling under the Sci/Tech class.

On the other hand, we can see a similar phenomenon happen with Sci/Tech articles. When inspecting Sci/Tech articles, it becomes evident that many are about Tech companies making progress on a project or seeing more / less sales. In other words, a lot of words relating to Business can be observed in Sci/Tech articles. The Business and Sci/Tech articles thus have some overlap, which causes misclassification among these categories. Looking back at the Sports category, it can be deduced that in Sports articles, a certain language is used that simply pertains mostly to only Sports articles, causing it to be less often confused or misclassified. Examples can include the name of a sport or sports team, sports terms like "Championship", "Grand prix" or sentences like "eight-run inning" and "A two-run deficit for a 4-2 win Sunday". The articles that do get misclassified are articles about, for example, lawsuits regarding Sports players, or news about the Olympics, which fall under the category World instead of Sports.
For the SVM, we can see the exact same phenomena: Sports is least often misclassified, Business and Sci/Tech are most often confused for each other, and Business is most often misclassified.

The division in which classes are often misclassified is very similar across both models; it seems that they make the same type of errors. A total of 24 cases of misclassification can be inspected in the Appendices for both the SVM (Appendix B, Table 3) and the Logistic Regression (Appendix C, Table 4). These tables categorise the errors based on their predicted and actual topic.

5.2 Specific Examples for Best Model

As the results show, the SVM is the better-performing model. Ten cases of misclassification for this model will be inspected in detail. The written-out text for these ten cases can be found in Appendix D (Table 6). It can be stated that the articles are misclassified because of specific word usage. A common occurrence is that articles belonging to the Business class talk about Scientific terms. This phenomenon can be observed in references 1 and 2. Reference 1 mentions "Web giant Yahoo" and "Web hosting products", which are common Tech terms; this article is misclassified as Sci/Tech. Reference 2 names a lot of different tech equipment; this article thus also gets misclassified as Sci/Tech. For reference 3 and 4, the misclassification can be seen quite easily; they are articles belonging to the World class, which talk about Olympic gold and a lawsuit regarding a Greek Weightlifter. The amount of Sports terms in these articles causes them to be classified as Sports articles. In references 5 and 6, the same thing happens for Business articles, where reference 5 mentions "Strategy" and "Staying ahead", while reference 6 plainly talks about a "Major League Baseball team", all of these are sports-related terms, causing a Sports classification. The inverse of this can be seen in references 7 and 8, where Sports articles regard Transfers or bids, which are business-related concepts. For reference 9, there are some Science terms being thrown around, like "energy" in combination with "fuel tank". For reference 10, it is a bit more difficult to detect which exact words cause the misclassification as Sci/Tech, but it might be words like "beams", "boost", "success" and "Stature in Europe" that are more often used in Sci/Tech articles than in Sports articles.

6 Discussion

The goal of this assignment was to discuss two different models, the SVM and Logistic Regression, in their capacity as potential baseline models for the AG News topic classification task. Based on a combination of the F1-score and the accuracy, the SVM is considered the superior baseline model as it slightly outperforms Logistic Regression in all metrics. Based on the confusion matrix and the more expanded error analysis, we can conclude that both models make very similar errors and are both valid baseline models. The errors can be summarised as being caused by terminologies being shared between different topics. For example, there are a lot of scientific terms used in Business and Sports articles.
There are also a few points of improvement. Firstly, there is the issue of normalisation. In our original pipeline, it was assumed that the TF-IDF feature was sufficiently normalised. In later testing, it seems that additional improvements could be made by normalising these values by using the standard deviation and assuming a zero mean (needed for sparse matrices). Lastly, significant improvements in training time and performance could be made by switching to the liblinear-based LinearSVC class, which is better optimised for large datasets [2]. This also uses a one-vs-one approach to multiclass classification, which seems to provide some small improvements as well. Due to time constraints, it was not possible to fully investigate the consequences of these changes for the error analysis, and this is left for future work.

References

[1] X. Zhang, J. Zhao, and Y. LeCun, "Character-level convolutional networks for text classification", in *Advances in Neural Information Processing Systems*, C. Cortes, N. Lawrence, D. Lee, M. Sugiyama, and R. Garnett, Eds., vol. 28, Curran Associates, Inc., 2015. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2015/file/250cf8b51c773f3f8dc8b4be867a9a02-Paper.pdf.

[2] F. Pedregosa et al., "Scikit-learn: Machine learning in Python", *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[3] K. P. Murphy, *Probabilistic Machine Learning: An introduction*. MIT Press, 2022. [Online]. Available: http://probml.github.io/book1.

[4] B. Schölkopf and A. J. Smola, *Learning with Kernels: Support Vector Machines, Regularization, Optimization, and Beyond*. The MIT Press, Dec. 2001, ISBN: 9780262256933. DOI: 10.7551/mitpress/4175.001.0001. [Online]. Available: https://doi.org/10.7551/mitpress/4175.001.0001.

A GitHub Repository

The URL to the GitHub Repository: https://github.com/TeunB6/natural-language-processing. For viewing purposes, the main branch is `main`.

B AI-Use Statement

Generative AI was utilised with the purpose and intention of accelerating the development of the project with regard to providing examples on implementing certain features, for example, the CLI UX. Permitted tools were OpenAI ChatGPT, Google Gemini, and GitHub Copilot, with the agreement that training of the models was disabled. Generated code was only used as a reference for comparison to the original written code, to provide any insight on technical scope and considerations we may have missed during development. Autocomplete features were used with human review, and the final product of the project is 100% human reviewed.

C Cases of missclassification from the SVM

Text	**Predicted**	**Actual**
Intel to delay product aimed for high-definition TVs SAN FRANCISCO – In the latest of a series of product delays, Intel Corp. has postponed the launch of a video display chip it had previously planned to introduce by year end, putting off a showdown with Texas Instruments Inc. in the fast-growing market for high-definition television displays.	Sci/Tech	Business
Yahoo! Ups Ante for Small Businesses Web giant Yahoo! is gambling that price cuts on its domain name registration and Web hosting products will make it more competitive with discounters in the space – which means that small businesses looking to move online get a sweeter deal through ...	Sci/Tech	Business
Teenage T. rex's monster growth Tyrannosaurus rex achieved its massive size due to an enormous growth spurt during its adolescent years.	Business	Sci/Tech
Storage, servers bruise HP earnings update Earnings per share rise compared with a year ago, but company misses analysts' expectations by a long shot.	Business	Sci/Tech
Venezuela Prepares for Chavez Recall Vote Supporters and rivals warn of possible fraud; government says Chavez's defeat could produce turmoil in world oil market.	Business	World
India's Tata expands regional footprint via NatSteel buyout (AFP) AFP - India's Tata Iron and Steel Company Ltd. took a strategic step to expand its Asian footprint with the announcement it will buy the Asia-Pacific steel operations of Singapore's NatSteel Ltd.	Business	World
SUVs: Live And Let Die NEW YORK - The newly released traffic crash fatality data have something for everyone in the debate about the safety of sport utility vehicles.	World	Business

Chavez victory confirmed Caracas, Venezuela - The results of an audit support the official vote count showing that President Hugo Chavez won this month #39;s recall referendum in Venezuela, the head of the Organization of American States said Saturday.	World	Business
Rocking the Cradle of Life When did life begin? One evidential clue stems from the fossil records in Western Australia, although whether these layered sediments are biological or chemical has spawned a spirited debate. Oxford researcher, Nicola McLoughlin, describes some of the issues in contention.	World	Sci/Tech
Atlantis "Evidence" Found in Spain, Ireland In 360 B.C. the Greek philosopher Plato described an island he called Atlantis. Now contradicting new evidence claims the fabled city-state was based on a real place.	World	Sci/Tech
Jackson Squares Off With Prosecutor SANTA MARIA, Calif. - Fans of Michael Jackson erupted in cheers Monday as the pop star emerged from a double-decker tour bus and went into court for a showdown with the prosecutor who has pursued him for years on child molestation charges...	Sports	World
Live: Olympics day four Richard Faulds and Stephen Parry are going for gold for Great Britain on day four in Athens.	Sports	World
Future Doctors, Crossing Borders Students at the Mount Sinai School of Medicine learn that diet and culture shape health in East Harlem.	Sci/Tech	World
Bryant Prosecutors Say Some Data Tainted DENVER - Crucial DNA evidence tested by defense experts in the Kobe Bryant sexual assault case might have been contaminated, prosecutors said in a court filing released Wednesday, just two days before jury selection is to begin. Prosecutors said they had found contamination in DNA "control" samples intended to ensure testing was accurate...	Sci/Tech	World
Prediction Unit Helps Forecast Wildfires (AP) AP - It's barely dawn when Mike Fitzpatrick starts his shift with a blur of colorful maps, figures and endless charts, but already he knows what the day will bring. Lightning will strike in places he expects. Winds will pick up, moist places will dry and flames will roar.	Sports	Sci/Tech
PRESS START FOR NOSTALGIA Like Led Zeppelin's "Stairway to Heaven"and Lynyrd Skynyrd's "Freebird", classic video games like Frogger and Pong can bring back an entire era.	Sports	Sci/Tech
Security scare as intruder dives in A CANADIAN husband's love for his wife has led to a tightening of security at all Olympic venues in Athens.	World	Sports
Don't expect Tiger to relinquish his top ranking without a fight They're calling Ohio a "battleground state", one of the two or three places likely to decide November's presidential election. On local TV, the Bush and Kerry ads air so frequently that it's easy to forget it's Bob Costas who actually runs the country.	World	Sports

For Craigslist, city was just the ticket Hadley Weinzierl used Craigslist to furnish her Jamaica Plain apartment, and when she bought a Maltese puppy, she sought advice from fellow Craigslisters on a good vet, a cheap dog-walker, and a park where she could let the dog run without a leash.	Sports	Business
Pen can pick some bike U-locks That U-shaped bike lock that you thought was so secure may be easy pickings for thieves who have nothing more sophisticated than a Bic pen.	Sports	Business
Tyncastle sale gets official go ahead Of the votes received by proxy and from shareholders in the room at a stormy extraordinary general meeting last night, 62.5 were in favour of the resolution.	Business	Sports
Icing call Out of money, out of patience, out of time, and for the foreseeable future, out of business.	Business	Sports
Olympics: Hansen still strong enough to take bronze Every ounce of his energy was expended, leaving an empty fuel tank. But, even in a depleted state, Brendan Hansen found a way to bolster his ever-growing swimming legacy.	Sci/Tech	Sports
Glory Comes Amid Empty Seats and Closed Shutters HERE in Old Europe, people install shutters outside their windows to keep out the heat, the pollution, the daylight, the noise. They also lock the shutters tight when they go away on holiday.	Sci/Tech	Sports

Table 3: 24 examples of misclassification within the SVM model

D Cases of misspecification from the Logistic Regression

Text	**Predicted**	**Actual**
Intel to delay product aimed for high-definition TVs SAN FRANCISCO – In the latest of a series of product delays, Intel Corp. has postponed the launch of a video display chip it had previously planned to introduce by year end, putting off a showdown with Texas Instruments Inc. in the fast-growing market for high-definition television displays.	Sci/Tech	Business
Yahoo! Ups Ante for Small Businesses Web giant Yahoo! is gambling that price cuts on its domain name registration and Web hosting products will make it more competitive with discounters in the space – which means that small businesses looking to move online get a sweeter deal through ...	Sci/Tech	Business
IBM to hire even more new workers By the end of the year, the computing giant plans to have its biggest headcount since 1991.	Business	Sci/Tech

Some People Not Eligible to Get in on Google IPO Google has billed its IPO as a way for everyday people to get in on the process, denying Wall Street the usual stranglehold it's had on IPOs. Public bidding, a minimum of just five shares, an open process with 28 underwriters - all this pointed to a new level of public participation. But this isn't the case.	Business	Sci/Tech
India's Tata expands regional footprint via NatSteel buyout (AFP) AFP - India's Tata Iron and Steel Company Ltd. took a strategic step to expand its Asian footprint with the announcement it will buy the Asia-Pacific steel operations of Singapore's NatSteel Ltd.	Business	World
Stocks Climb on Drop in Consumer Prices NEW YORK - Stocks rose for a second straight session Tuesday as a drop in consumer prices allowed investors to put aside worries about inflation, at least for the short term. With gasoline prices falling to eight-month lows, the Consumer Price Index registered a small drop in July, giving consumers a respite from soaring energy prices...	Business	World
Oil Hits New High Over $48 as Iraq Violence Flares LONDON (Reuters) - Oil prices struck a fresh record above $48 a barrel on Thursday, spurred higher by renewed violence in Iraq and fresh evidence that strong demand growth in China and India has not been slowed yet by higher energy costs.	World	Business
Chavez victory confirmed Caracas, Venezuela - The results of an audit support the official vote count showing that President Hugo Chavez won this month's recall referendum in Venezuela, the head of the Organization of American States said Saturday.	World	Business
Iran shuts reformist websites WEBSITES CLOSE to Iran's leading reformist party have been blocked by religious hardliners in the police bureau of public morals.	World	Sci/Tech
U.N. Urges Funds to Curb African Locusts (AP) AP - With swarms of locusts threatening crops in a number of African countries, a U.N. agency appealed for an additional 70 million in assistance Thursday to prevent the upsurge from becoming a full-scale plague.	World	Sci/Tech
Google Lowers Its IPO Price Range SAN JOSE, Calif. - In a sign that Google Inc.'s initial public offering isn't as popular as expected, the company lowered its estimated price range to between $85 and $95 per share, down from the earlier prediction of $108 and $135 per share...	Sci/Tech	World
Future Doctors, Crossing Borders Students at the Mount Sinai School of Medicine learn that diet and culture shape health in East Harlem.	Sci/Tech	World

Jackson Squares Off With Prosecutor SANTA MARIA, Calif. - Fans of Michael Jackson erupted in cheers Monday as the pop star emerged from a double-decker tour bus and went into court for a showdown with the prosecutor who has pursued him for years on child molestation charges...	Sports	World
Live: Olympics day four Richard Faulds and Stephen Parry are going for gold for Great Britain on day four in Athens.	Sports	World
Prediction Unit Helps Forecast Wildfires (AP) AP - It's barely dawn when Mike Fitzpatrick starts his shift with a blur of colorful maps, figures and endless charts, but already he knows what the day will bring. Lightning will strike in places he expects. Winds will pick up, moist places will dry and flames will roar.	Sports	Sci/Tech
PRESS START FOR NOSTALGIA Like Led Zeppelin's 'Stairway to Heaven' and Lynyrd Skynyrd's 'Freebird' classic video games like Frogger and Pong can bring back an entire era.	Sports	Sci/Tech
Security scare as intruder dives in A CANADIAN husband's love for his wife has led to a tightening of security at all Olympic venues in Athens.	World	Sports
Careening Indians Fall Slumping Cleveland lost a three-run lead while Derek Jeter homered and stole two ninth-inning bases as New York sent the Indians to their ninth consecutive loss, 5-4, Tuesday.	World	Sports
For Craigslist, city was just the ticket Hadley Weinzierl used Craigslist to furnish her Jamaica Plain apartment, and when she bought a Maltese puppy, she sought advice from fellow Craigslisters on a good vet, a cheap dog-walker, and a park where she could let the dog run without a leash.	Sports	Business
Pen can pick some bike U-locks That U-shaped bike lock that you thought was so secure may be easy pickings for thieves who have nothing more sophisticated than a Bic pen.	Sports	Business
Bonds's Excuse Has the Scent of Snake Oil, Not Arthritis Balm Barry Bonds's ignorance of the substances obtained by his trainer from Balco is more childish than a youngster's excuse that "the dog ate my homework."	Business	Sports
Tyncastle sale gets official go ahead Of the votes received by proxy and from shareholders in the room at a stormy extraordinary general meeting last night, 62.5 were in favour of the resolution.	Business	Sports
Olympics: Hansen still strong enough to take bronze Every ounce of his energy was expended, leaving an empty fuel tank. But, even in a depleted state, Brendan Hansen found a way to bolster his ever-growing swimming legacy.	Sci/Tech	Sports
Sprint is chock full of potential heros It would be nice to see this week's 100-meter sprint as simply the best footrace of all time. We could witness four sub-10-second sprints for the first time ever. It would be nice to watch with raised eyebrows instead of furrowed ones. It ...	Sci/Tech	Sports

Table 4: 24 examples of misclassification within the Logistic regression model

E Ten specific cases of misclassification from the SVM

Ref	Text	Predicted	Actual
1	Yahoo! Ups Ante for Small Businesses Web giant Yahoo! is gambling that price cuts on its domain name registration and Web hosting products will make it more competitive with discounters in the space – which means that small businesses looking to move online get a sweeter deal through ...	Sci/Tech	Business
2	HP to tempt holiday shoppers with sights and sounds The computer-hardware giant, best known for products such as PCs and printers, on Friday laid out its plan to become a brand-name in consumer electronics products such as flat-screen TVs, music players and the devices that move content between them.	Sci/Tech	Business
3	Live: Olympics day four Richard Faulds and Stephen Parry are going for gold for Great Britain on day four in Athens.	Sports	World
4	Greek weightlifter awaits verdict Greek weightlifter Leonidas Sampanis will find out on Sunday if he is to be stripped of his medal.	Sports	World
5	Hasbro's No Conehead Its Saturday Night Live version of "Trivial Pursuit" is good strategy for staying ahead of age compression.	Sports	Business
6	Skyboxes May Reach A Limit Verizon Communications Inc. would seem to be a prime candidate to snap up a luxury suite at the $440 million ballpark being planned for Washington's Major League Baseball team.	Sports	Business
7	City Series-ly wounded With the Yanks out of the World Series, the city's economy loses out on at least $40 million, according to studies by the controller's office and other city agencies.	Business	Sports
8	Cape Town F1 bid And they plan to build a multi-million-dollar race track here for the event, an official said today. quot;The plans for the bid are far advanced.	Business	Sports
9	Olympics: Hansen still strong enough to take bronze Every ounce of his energy was expended, leaving an empty fuel tank. But, even in a depleted state, Brendan Hansen found a way to bolster his ever-growing swimming legacy.	Sci/Tech	Sports
10	A 'new Greece' beams after success of Games As Greeks get a boost, it remains unclear if success will mean higher stature in Europe. By Peter Ford Staff writer of The Christian Science Monitor.	Sci/Tech	Sports

Table 6: Ten cases of misclassification from the SVM

F Gridsearch

C	Mean Test Score	Std Test Score
0.001	0.8130	0.0103
0.01	0.8900	0.0050
0.1	**0.9021**	0.0018
1.0	0.9015	0.0031
10.0	0.8716	0.0057
100.0	0.8112	0.0062
1000.0	0.8071	0.0080

Table 7: The table above showcases the results of the gridsearch. The test score used was the accuracy; the values were computed on a 5-fold cross-validation. The maximum test score was found for $C = 0.1$ and is presented in bold.